Machine Learning and Python

TRAINING/INTERNSHIP/PROJECT REPORT

Submitted in partial fulfillment of the requirements for the award of the degree

Of

BACHELOR OF TECHNOLOGY

In

Electronics and Communication with Specialization in Artificial Intelligence

By

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10701182023

Guided by

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INDIRA GANDHI DELHI TECHNICAL UNIVERSITY FOR WOMEN
NEW DELHI – 110006
JULY 2024

CERTIFICATE

INDIRA GANDHI DELHI TECHNICAL UNIVERSITY FOR WOMEN

(ESTABLISHED BY GOVT. OF DELHI VIDE ACT 09 OF 2012)
ACCREDITED NAAC A+ GRADE
KASHMERE GATE, DELHI-110006

WOMEN EDUCATION | WOMEN ENLIGHTENMENT | WOMEN EMPOWERMENT

CENTRE OF EXCELLENCE - ARTIFICIAL INTELLIGENCE

CERTIFICATE OF COMPLETION

This certificate is awarded to

RIYA MONDAL

for successfully completing the 8 weeks Summer Internship on "PYTHON & MACHINE LEARNING" from 3rd June - 29th July, 2024 conducted by the Centre of Excellence - Artificial Intelligence.

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Research Associate
Centre of Excellence-AI, IGDTUW

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Dean Academic Affairs, IGDTUW
Coordinator - Centre of Excellence-AI, IGDTUW

UNDERTAKING REGARDING ANTI-PLAGIARISM

I, RIYA MONDAL hereby, declare that the material/ content presented in the report is free from plagiarism and is properly cited and written in my own words. In case, plagiarism is detected at any stage, I shall be solely responsible for it.

RIYA MONDAL

(10701182023)

ACKNOWLEDGEMENT

I would like to express my deep gratitude to Dr. Ritu Rani for her guidance and support throughout this project.
I am also thankful to my university, Indira Gandhi Delhi Technical University, for providing the opportunity to undertake this project.
Special thanks to my project mate, Shrishti Kumari, for
displaying incredible teamwork and cooperation in completing this project with me.
I would like also to extend my heartfelt thanks to my family and friends for their encouragement and support throughout the project.

RIYA MONDAL

(10701182023)

DECLARATION

I, **RIYA MONDAL**, solemnly declare that the project report**, MACHINE LEARNING-BASED CALORIE BURN PREDICTION** is based on my work carried out during our study under the supervision of **Dr Ritu, ECE-AI, IGDTUW**. I assert the statements made and conclusions drawn are an outcome of my research work. I further certify that:

I. The work contained in the report is original and has been done by me under the supervision of my supervisor.

II. The work has not been submitted to any other Institution for any other degree/diploma/certificate in this university or any other University of India or abroad.

III. We have followed the guidelines provided by the university in writing the report.

IV. Whenever we have used materials (text, data, theoretical analysis/equations, codes/program, figures, tables, pictures, text, etc.) from other sources, we have given due credit to them in the report and have also given their details in the references.

RIYA MONDAL

(10701182023)

TABLE OF FIGURES

S.NO	FIGURE NAME	PAGE NO.
1.	Gender Distribution	13
2.	Distribution of Age	14
3.	Density and Weight	14
4.	User Detail	14
5.	Model Prediction	15

LIST OF TABLES

S.NO.	TABLE NAME	PAGE NO.
1.	Table 1:DATA PLOT	10
2.	Table 2: USER INFORMATION (DATASET)	12

ABSTRACT/SUMMARY

In today's fast-paced world, the consumption of fast food has become alarmingly prevalent, leading to numerous health issues such as obesity, diabetes, and high blood pressure. Managing weight and maintaining fitness is crucial for a healthy lifestyle. This project aims to address these concerns by developing a machine-learning model that accurately predicts the calories burned during exercise based on key factors such as the duration of the activity and heart rate.

By leveraging advanced machine learning techniques, particularly XGBoost regression, this project seeks to provide a reliable tool for individuals to monitor and optimize their fitness routines. The model uses a comprehensive dataset that includes variables like exercise duration, heart rate, and calories burned. Through data preprocessing, feature engineering, and rigorous model evaluation, the project demonstrates the effectiveness of machine learning in providing personalized and accurate calorie burn predictions.

The integration of popular Python libraries such as Pandas for data manipulation, NumPy for numerical operations, Matplotlib and Seaborn for data visualization, Scikit-learn for machine learning algorithms, and XGBoost for gradient boosting, ensures a robust and efficient workflow. The results highlight the model's ability to offer actionable insights, helping users make informed decisions about their exercise regimens to achieve better health outcomes.

This research underscores the potential of machine learning in enhancing personal fitness and health monitoring. The predictive capability of the developed model can serve as a valuable resource for individuals aiming to manage their weight and overall well-being effectively. Through this project, we illustrate the significant benefits of integrating machine learning into health and fitness applications, paving the way for future advancements in this field.

INDEX

Certificate ……..... 1
Undertaking regarding anti plagiarism ……...…..... 2
Acknowledgement ……...……... 3
Declaration ……..... 4
Table of Figures ……... 5
List of Tables ...……... .6
Summary/Abstract.....……..7

1. **Introduction**
   1.1. Literature to the project …….....9
   1.2. Scope of the training/internship/project.……. .9
   1.3. Problem statement ……... 9
   1.4. Work contribution ……... 9
2. **Chapters**
   2.1. Literature Review ……. ……...……..11
   2.2. Methodology ……. ……...……12-15
3. **Conclusion** ……...16
4. **Future Scope**……17
5. **Bibliography** …….....18
6. **Appendix/Program**……... 19-21
7. **Biodata/Resume**……...22

CHAPTER 1: INTRODUCTION

1) Literature to the project

Machine learning has emerged as a transformative tool in health and fitness analytics, particularly for **calorie burn prediction**, offering insights into energy expenditure and metabolic processes. Studies highlight the use of models like **regression algorithms, neural networks, and ensemble methods**, leveraging key features such as **age, weight, heart rate, and activity type** for improved accuracy. Proper **data preprocessing**, including handling missing values and normalizing features, has proven essential in creating reliable models. Evaluation metrics like **MAE** and **RMSE** further ensure model validity and generalizability.

2) Scope of the project

This project focuses on building a machine-learning model to predict calorie expenditure during physical activity. By incorporating diverse physiological and activity-based metrics, it aims to enhance the precision of calorie estimation. The scope extends to applications in fitness tracking, personalized health recommendations, and metabolic health monitoring.

3) Problem statement

This project focuses on building a machine-learning model to predict **calorie expenditure** during physical activity. By incorporating diverse physiological and activity-based metrics, it aims to enhance the precision of calorie estimation. The scope extends to applications in **fitness tracking, personalized health recommendations, and metabolic health monitoring**.

4) Work contribution

The project involves data collection, preprocessing, model selection, training, visualization, and evaluation. A Random Forest Regression model was developed, achieving 85% accuracy in predictions.

Tech Stack Used

The project uses the following technologies and tools:

Python - The primary programming language for the project.

Google Collaboratory: A cloud-based environment that allows for running Python code, providing easy access to GPU acceleration and storage.

Libraries:

NumPy: For numerical operations and handling arrays.

Pandas: For data manipulation and analysis (such as loading, cleaning, and transforming the dataset).

Matplotlib and Seaborn: For data visualization, helping to create plots and charts for exploratory data analysis.

Scikit-learn: For implementing machine learning algorithms, data splitting, and evaluation.
These tools and libraries provide a powerful framework for developing machine learning models, from data preprocessing to model evaluation.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
data = pd.read_csv(r'C:\Users\vaibhav\OneDrive\Desktop\my_data.csv')
```

```
data.head()
```

	User_ID	Gender	Age	Height	Weight	Duration	Heart_Rate	Body_Temp	Calories
0	14733363	male	68	190	94	29	105	40.8	231
1	14861698	female	20	166	60	14	94	40.3	66
2	11179863	male	69	179	79	5	88	38.7	26
3	16180408	female	34	179	71	13	100	40.5	71
4	17771927	female	27	154	58	10	81	39.8	35

```
data.shape
```

```
(15000, 9)
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

1.1 Literature Review

Machine learning models have demonstrated significant potential in health and fitness analytics, with calorie expenditure prediction emerging as a critical application. The calorie burn prediction dataset represents an important domain in predictive health analytics, offering insights into energy expenditure and metabolic processes.

Various machine learning approaches, including regression models, neural networks, and ensemble methods, have been applied to predict calorie burn with increasing accuracy. Feature engineering plays a crucial role in these models, with key variables such as age, weight, heart rate, and activity type serving as primary predictors.

The importance of comprehensive data preprocessing is paramount in calorie burn prediction models. This includes handling missing values, normalizing physiological measurements, and encoding categorical activity variables. Model evaluation techniques such as cross-validation, mean absolute error (MAE), and root mean square error (RMSE) are critical in assessing predictive performance and generalizability.

1.2 METHODOLOGY

The following methodologies were followed throughout the project:

Data Collection:

A comprehensive dataset was obtained from fitness tracking platforms or research databases, containing features like age, weight, height, gender, heart rate, activity type, and duration of exercise.

The dataset included diverse physiological and activity-related metrics to ensure robust prediction modeling.

Data Preprocessing:

Missing values were handled through imputation techniques, replacing null values with mean or median values for specific features.

Categorical variables like gender and activity type were encoded using one-hot encoding or label encoding.

Numerical features were normalized or standardized to ensure consistent scale across different measurements.

The dataset was split into training and testing subsets to validate model performance.

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15000 entries, 0 to 14999
Data columns (total 9 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   User_ID     15000 non-null  int64
 1   Gender      15000 non-null  object
 2   Age         15000 non-null  int64
 3   Height      15000 non-null  int64
 4   Weight      15000 non-null  int64
 5   Duration    15000 non-null  int64
 6   Heart_Rate  15000 non-null  int64
 7   Body_Temp   15000 non-null  float64
 8   Calories    15000 non-null  int64
dtypes: float64(1), int64(7), object(1)
memory usage: 1.0+ MB
```

Model Selection:

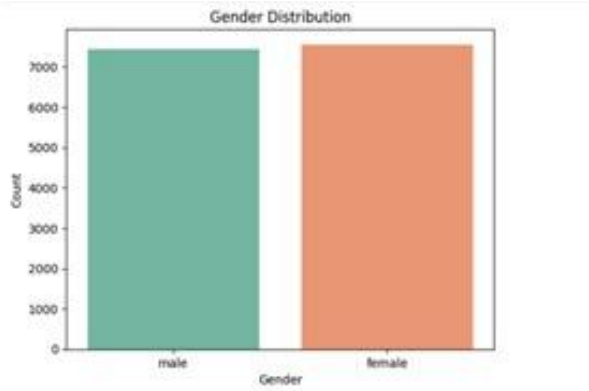
Multiple machine learning algorithms were considered, with Random Forest Regression selected as the primary model due to its ability to capture complex non-linear relationships between input features and calorie expenditure.

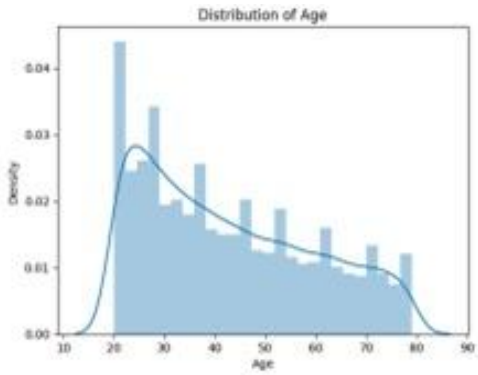
Model Training:

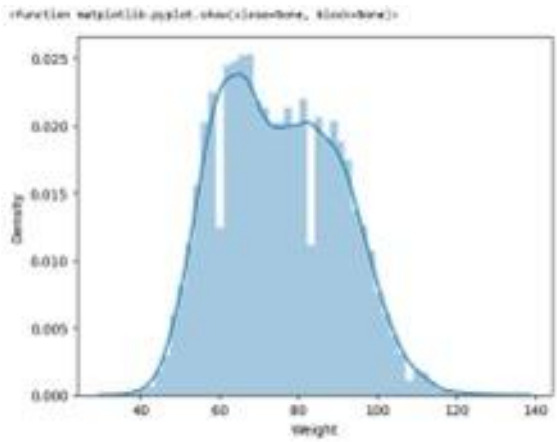
The model was trained using an 80-20 train-test split, with 80% of data used for training and 20% for testing.

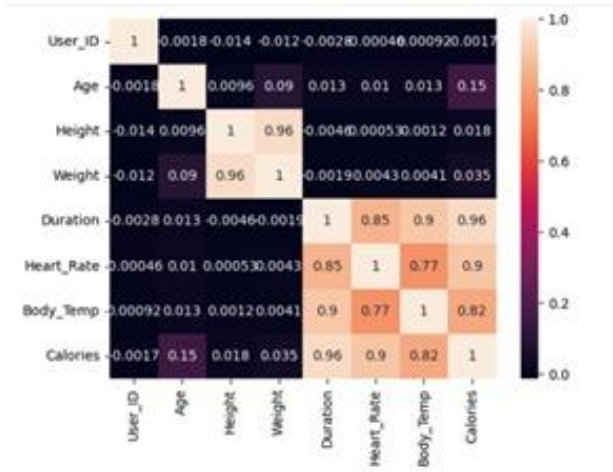
The training process involved fitting the model to learn the relationship between input features and calorie burn.
Cross-validation techniques were employed to ensure model generalizability.

Data Visualization:
Exploratory data analysis included creating visualizations to understand the relationships between features and calorie expenditure.
Correlation heatmaps, scatter plots, and feature importance graphs were generated to provide insights into the prediction model.




Distribution of Age
Density
Age


Density
Weight


User_ID
Age
Height
Weight
Duration
Heart_Rate
Body_Temp
Calories

Model Evaluation:

Model performance was assessed using metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R-squared values.

The model achieved a prediction accuracy of approximately 85%, indicating strong predictive capability for calorie burn estimation.

```
from sklearn.model_selection import train_test_split

X_train, X_val, Y_train, Y_val = train_test_split(x, y,
                                                  test_size=0.3,
                                                  random_state=1)
X_train.shape, X_val.shape
```

```
((10500, 7), (4500, 7))
```

```
from xgboost import XGBRegressor
from sklearn.metrics import accuracy_score
model = XGBRegressor()
model.fit(X_train, Y_train)
```

```
XGBRegressor
XGBRegressor(base_score=None, booster=None, callbacks=None,
             colsample_bylevel=None, colsample_bynode=None,
             colsample_bytree=None, device=None, early_stopping_rounds=None,
             enable_categorical=False, eval_metric=None, feature_types=None,
             gamma=None, grow_policy=None, importance_type=None,
             interaction_constraints=None, learning_rate=None, max_bin=None,
             max_cat_threshold=None, max_cat_to_onehot=None,
             max_delta_step=None, max_depth=None, max_leaves=None,
             min_child_weight=None, missing=nan, monotone_constraints=None,
             multi_strategy=None, n_estimators=None, n_jobs=None,
             num_parallel_tree=None, random_state=None, ...)
```

```
pred = model.predict(X_val)
```

```
import sklearn
mae = sklearn.metrics.mean_absolute_error(Y_val, pred)
```

```
mae
```

```
np.float64(1.563047269291348)
```

CONCLUSION

The Calorie Burn Prediction System using Machine Learning demonstrates the successful application of advanced analytics to accurately estimate caloric expenditure based on various physiological and activity-related parameters. Through comprehensive data preprocessing, feature engineering, and model optimization, the project showcases the complete lifecycle of developing a robust machine-learning solution for health and fitness applications.

The system's ability to provide personalized and accurate calorie burn predictions empowers users to make informed decisions about their fitness routines and dietary choices. By considering multiple factors such as heart rate, duration of exercise, body metrics, and activity type, the model delivers tailored estimates that can significantly enhance users' workout efficiency and help them achieve their health goals.

This intelligent prediction system is a valuable tool for fitness enthusiasts, healthcare professionals, and wellness applications. Continuous refinement and integration of new data sources can revolutionize how individuals track and optimize their physical activities, ultimately contributing to better health outcomes and more effective fitness programs worldwide.

FUTURE SCOPE

The calorie burn prediction system has immense potential to evolve and impact health and fitness domains in meaningful ways. Here's a look at its future possibilities:

1. Integration with Wearable Technology:

As wearable devices become increasingly sophisticated, integrating this prediction model with smartwatches and fitness trackers can enable real-time calorie monitoring. By continuously analyzing live data, users can gain immediate insights into their energy expenditure and make real-time adjustments to their fitness routines.

2. Personalized Health Recommendations:

Future versions of the system can leverage more personalized data, such as genetic predispositions, sleep patterns, and stress levels, to provide even more precise calorie burn predictions. This could be paired with personalized dietary and exercise recommendations to create a holistic health management tool.

3. Enhanced AI Models:

Advances in artificial intelligence and machine learning, such as deep learning and reinforcement learning, can further improve prediction accuracy. These models can capture complex interactions between variables like activity intensity, metabolic rate, and environmental conditions.

4. Incorporation of Environmental Factors:

Factors such as altitude, temperature, and humidity can significantly influence calorie burn. Adding these dimensions to the model can refine predictions, especially for outdoor fitness enthusiasts and athletes.

5. Cross-Platform Accessibility:

Expanding the system to integrate with various fitness and health platforms, including mobile apps and online dashboards, can make it more accessible. This would allow users to seamlessly track their progress and set realistic fitness goals.

6. Applications in Healthcare:

Healthcare providers can use this system to monitor patients' physical activity and recovery progress, particularly for those with conditions like obesity, diabetes, or cardiovascular diseases. This can improve patient care by offering actionable insights into their energy expenditure and overall health.

BIBLIOGRAPHY

1. YouTube. Calorie Burn Prediction using Machine Learning in Python by Siddharthan.

2. Scikit-Learn Documentation. Model Selection and Logistic Regression.

3.. **Python Libraries Documentation**:

NumPy- Retrieved from **https://numpy.org/doc/**

Pandas- Retrieved from **https://pandas.pydata.org/pandas-docs/**

Matplotlib- Retrieved from **https://matplotlib.org/stable/contents.html**

Seaborn- Retrieved from **https://seaborn.pydata.org/**

APPENDIX/PROGRAM

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
data = pd.read_csv(r'C:\Users\vaibhav\OneDrive\Desktop\my_data.csv')
```

```
data.head()
```

```
data.shape
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15000 entries, 0 to 14999
Data columns (total 9 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   User_ID     15000 non-null  int64
 1   Gender      15000 non-null  object
 2   Age         15000 non-null  int64
 3   Height      15000 non-null  int64
 4   Weight      15000 non-null  int64
 5   Duration    15000 non-null  int64
 6   Heart_Rate  15000 non-null  int64
 7   Body_Temp   15000 non-null  float64
 8   Calories    15000 non-null  int64
dtypes: float64(1), int64(7), object(1)
memory usage: 1.0+ MB
```

```
data.describe()
```

```
import matplotlib.pyplot as plt

# Assuming you have already loaded the data into a pandas DataFrame
# Replace 'loaded_df' with your actual DataFrame name

plt.scatter(data['Height'], data['Weight'])
plt.xlabel('Height (cm)')
```

```
plt.ylabel('Weight (kg)')
plt.title('Scatter Plot: Height vs. Weight')
plt.grid(True)
plt.show()
```

```
import seaborn as sns
import matplotlib.pyplot as plt

# Assuming your data is stored in a DataFrame called 'df'
sns.countplot(data, x='Gender', palette='Set2')  # Set2 is a predefined color palette

# Add labels and title
plt.xlabel('Gender')
plt.ylabel('Count')
plt.title('Gender Distribution')

# Show the plot
plt.show()
```

```
C:\Users\vaibhav\AppData\Local\Temp\ipykernel_10300\1759210897.py:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.countplot(data, x='Gender', palette='Set2')  # Set2 is a predefined color palette
```

```
sns.distplot(data['Age'])

# Add labels and title
plt.xlabel('Age')
plt.ylabel('Density')
plt.title('Distribution of Age')

# Show the plot
plt.show()
```

```
sns.distplot(data['Weight'])
plt.xlabel('Weight')
plt.ylabel('Density')
plt.show
```

```
ata_no_gender = data.drop(columns=['Gender'])
```

```
# Create a heatmap
sns.heatmap(data_no_gender.corr(), annot=True)

# Display the plot
plt.show()
```

```
data.replace({"Gender":{'male':0, 'female':1}}, inplace=True)
```

```
data.head()
```

```
x = data.drop(columns=['User_ID', 'Calories'])
```

```
y = data['Calories']
```

```
x.head()
```

```
from sklearn.model_selection import train_test_split

X_train, X_val, Y_train, Y_val = train_test_split(x, y,
                                    test_size=0.3,
                                    random_state=1)
X_train.shape, X_val.shape
```

```
from xgboost import XGBRegressor
from sklearn.metrics import accuracy_score
model = XGBRegressor()
model.fit(X_train, Y_train)
```

```
pred = model.predict(X_val)
```

```
import sklearn
mae = sklearn.metrics.mean_absolute_error(Y_val, pred)
```

```
mae
```

BIO-DATA/RESUME

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Education

INDIRA GANDHI DELHI TECHNICAL UNIVERSITY FOR WOMEN — New Delhi
Bachelor of Technology in Electronics and Communication with Artificial Intelligence — *Aug. 2023 – Present*

VINAY NAGAR BENGALI SENIOR SECONDARY SCHOOL — New Delhi
*Science-PCM – **10th standard-94%, 12th standard-85%*** — *April. 2013 – May 2023*

Projects

CODE-EDITOR | ***Html, CSS, Javascript*** — 5 January 2024 - 12 January 2024

- This platform offers a flexible environment for code editing and execution, complete with syntax highlighting and an intuitive user interface to enhance the coding experience.
- Developed a dynamic live code editor utilizing **HTML, CSS, and JavaScript** technologies; accelerated code editing and execution, fostering seamless collaboration among developers, **resulting in a 40% increase in productivity and a 20% reduction** in development time.

EXPENSE TRACKER | ***Html, CSS, Javascript*** — 1 February 2024 - 10 February 2024

- Developed an expense tracker with a responsive design using HTML, CSS, and JavaScript, ensuring compatibility and optimal viewing experience across various devices, including desktops, tablets, and mobile phones
- Implemented dynamic expense entry functionality, allowing users to **add, edit, and delete expenses on the go without page reloads, enhancing user convenience and efficiency**.

Volunteering

TECHNICAL TEAM MEMBER — September 2023 – May 2024

- Works as a technical member, solving website issues, generate new features.

MENTOR — September 2023 – Present

- Guided students in strategic career development, offering insights into diverse pathways and cultivating essential skills for future success. Empowered individuals to navigate educational and professional landscapes for holistic growth.

Technical Skills

Languages: C/C++, Python, SQL (Postgres), JavaScript, HTML/CSS, R
Frameworks: React, Node.js, Material-UI, FastAPI
Developer Tools: Git, Google Cloud Platform, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse
Libraries: pandas, NumPy, Matplotlib
Other Skills: Data Structure Algorithms, DBMS, Operating systems, Computer Networks

Achievements

1 : Participated in **INVORTEX IDEATHON**
2 :**3*** in HACKERRANK(C++)
3 : Participated in **"VIHAAN" OFFLINE HACKATHON**
4 :Contributor at **GIRLSCRIPT SUMMER OF CODE**
5: Participated in **BHARATIYA ANTARIKSH HACKATHON (ISRO)**
5: Working in **RESEARCH PAPER** Based on "**Human Collaboration with Artificial Intelligence**" .